MILITARY SPECIFICATION

LOUDSPEAKER, PERMANENT MAGNET, LS-474/U

1. SCOPE

1.1 This specification covers the general requirements of a 4 watt universal communications loudspeaker for use with electronic equipments.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-27 - Transformers and Inductors, Audio, Power and High Power Pulse, General Specification for.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and System, Requirements for.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals).
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-W-16878 - Wire, Electrical, Insulated, High Temperature.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

BUREAU OF SHIPS

RE49F767 - Loudspeaker, Permanent Magnet, LS-474/U.
RE49C764 - Loudspeaker, Permanent Magnet, NT491814.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

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3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.2).

3.2 Material. - The material for each part shall meet the requirements of this specification. When a material is not specified, it shall be selected from MIL-E-16400.

3.3 Soldering. - All soldering shall be in accordance with MIL-E-16400.

3.4 Mechanical. -

3.4.1 Machine screws. - All machine screws used throughout the loudspeaker shall be corrosion-resistant. All screws shall employ "PHILLIPS" head or equivalent screwdriver slots.

3.4.2 Dimensions. - The dimensions and external physical characteristics of the loudspeaker shall be in accordance with Drawing RE49F767.

3.4.3 Weight. - The total weight of the loudspeaker including the mounting brackets shall not exceed 10 pounds.

3.4.4 Enclosure. - The loudspeaker shall be totally enclosed in accordance with MIL-STD-108.

3.4.5 Cabinet construction. - The cabinet shall be fabricated in accordance with Drawing RE49F767.

3.4.6 Front panel. - The front panel shall contain and support all electrical parts. The terminal block may be mounted on the cabinet or the front panel. The loudspeaker shall operate through an opening in the front panel. This opening shall be covered by a perforated metal screen to protect the loudspeaker cone.

3.4.7 Cable entrance. - The loudspeaker shall be furnished with a grommet to accommodate a 3/8 inch cable. It shall be possible to provide cable entrance from the top, bottom, or back of the cabinet. A cable clamp shall also be provided on the interior of the cabinet for use in securing the cable prior to installation.

3.4.8 Terminal block. - The terminal block shall be marked to indicate the proper match to the external audio input line in accordance with Drawing RE49F767.

3.5 Electrical. -

3.5.1 Circuit. - The circuit of the loudspeaker shall be as specified on Drawing RE49F767.

3.5.2 Hook-up wire. - All hook-up wire shall be in accordance with type B of MIL-W-16878.

3.5.3 Matching transformer. - The matching transformer may be shaped round or rectangular and shall be provided with a 600 ohm ± 10 percent primary and a 3.2 ohm ± 10 percent secondary winding. The transformer shall meet the requirements of grade 4 or 5, class R, life expectancy X of MIL-T-27.

3.5.3.1 Transformer marking. - The transformer shall be marked with the manufacturer's prefix and Federal stock number in a location not obstructed from view. The markings shall be legible for the normal life of the transformer.

3.5.4 Loudspeaker unit. - The loudspeaker unit shall be of the permanent magnet type, in accordance with Drawing RE49C764.

3.5.5 Power. - Through the frequency range of 200 to 5,000 cycles per second (c.p.s.), the loudspeaker shall be capable of handling a constant voltage having a root mean square (r.m.s.) value to provide 4 watts of power at 400 c.p.s.

3.5.6 Frequency response. - The loudspeaker shall produce articulately and audibly, frequencies within the range of 200 to 5,000 c.p.s. The frequency response shall be measured with a voltage having a constant r.m.s. value, which will provide 4 watts of power at 400 c.p.s. plus or minus 20 percent, applied across the

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600-ohm input of the matching transformer. The frequency response of the loudspeaker assembly shall not vary more than ± 6 db from the limits indicated in table I. Because of mechanical limitations of dimensions, it is not required that the response of the loudspeaker be essentially flat throughout the specified range, but only that there be no serious response peaks of such amplitude as to affect materially the quality above 200 c.p.s.

Table I - Frequency response.

Frequency	Output (See 4.4.1(g))
c.p.s.	Db
200 to 1,000	97
1,000 to 5,000	99

3.5.7 Variation in a.f. response. - The variation at any frequency between 200 and 5,000 c.p.s. shall be not greater than 3 db from the value measured prior to the environmental tests.

3.5.8 Harmonic distortion. The total harmonic distortion in the acoustic output of the loudspeaker assembly shall not exceed 7 percent of the input signals when tested as specified in 4.4.8. After the environmental tests, the distortion shall not exceed the value specified herein.

3.5.9 Acoustical efficiency. - The acoustical efficiency of the loudspeaker shall be tested with a constant input of 4 watts applied to the voice coil within the frequency range of 700 to 1,300 c.p.s. The acoustical pressure shall average at least 20 dynes per square centimeter along the speaker axis. The measurements shall be made in accordance with 4.4.1, in an acoustically deadened room.

3.5.10 Acoustic quality. - An acoustic quality test shall be performed on each loudspeaker. There shall not be any buzzing, rattling, or other spurious sounds in the output of the loudspeaker assembly which will degrade the quality of reproduced speech.

3.6. Environmental. -

3.6.1 Requirements after environmental tests. - After the environmental tests, the loudspeaker shall show no evidence of mechanical or electrical failures. There shall be no signs of peeling or flaking of the painted finish, or corroding of the metallic parts. There shall be no evidence of loudspeaker cone deformation.

3.6.1.1 After the environmental tests specified in 4.4.6, 4.4.9, 4.4.10 and 4.4.11, the loudspeaker shall meet the requirements of 3.5.7, 3.5.8 and 3.5.10.

3.6.2 Shock and vibration. - The loudspeaker shall be designed to withstand the mechanical shocks and vibrations encountered aboard ship when tested as specified in 4.4.9.

3.6.3 Dielectric strength. - There shall be no evidence of arcing or insulation breakdown when the loudspeaker is tested as specified in 4.4.5.

3.6.4 Temperature cycling. - There shall be no malfunction of the loudspeaker when subjected to the test specified in 4.4.10.

3.6.5 Moisture resistance. - The loudspeaker shall be tested as specified in 4.4.11 to determine conformance with this specification.

3.6.6 Salt spray. - The loudspeaker shall withstand the salt spray test specified in 4.4.6. Following the salt spray test, the requirements of 3.6.1 and 3.6.1.1 shall be met.

3.7 Identification plates. - Identification plates shall be in accordance with Drawing RE49F767.

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3.8 Painting. - The external and internal surfaces shall have all rust or other visible corrosion products and flux removed; shall be thoroughly cleaned of all grease, oil, and dirt by solvent wiping, vapor degreasing or caustic washing and rinsing; and shall then be painted as follows:

(a) Primer - One coat of wash pretreatment in accordance with MIL-P-15328, applied as a continuous film 0.0002 to 0.0005 inch thick.
(b) Enamel - Two coats of gray enamel, conforming to class 2 of MIL-E-15090, shall be applied as continuous films approximately 0.001 inch thick.

3.9 Workmanship. - Workmanship shall be such that the finished product will meet all requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification; where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - Unless otherwise specified (see 6.2), two preproduction samples shall be made and assembled by the manufacturer with tools and methods, that, as far as practicable, are the same as those which will be used for quantity production of the product. The manufacture of the loudspeaker for a Government contract, prior to approval of the preproduction sample, shall be at the manufacturer's risk.

4.2.1 Test routine. - The preproduction samples shall be subjected to the tests specified in table II in the order listed. No failures will be permitted. When the loudspeaker unit NT491814 is provided as part of an assembly, testing before assembly is not required.

Table II - Preproduction inspection.

Examination and test	Requirement paragraph	Test paragraph
Visual, dimensional and mechanical examination	3.4.1 thru 3.4.4, 3.7, 3.8 and 3.9	4.4.3
Power handling	3.5.5	4.4.4
Dielectric strength	3.6.3	4.4.5
Frequency response	3.5.6	4.4.4
Acoustical quality	3.5.10	4.4.7
Acoustical efficiency	3.5.9	4.4.4
Harmonic distortion	3.5.8	4.4.8
Temperature cycling	3.6.4	4.4.10
Moisture resistance	3.6.5	4.4.11
Shock and vibration	3.6.2	4.4.9
Salt spray	3.6.6	4.4.6

4.3 Quality conformance inspection. -

4.3.1 Lot. - All loudspeakers offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.3.2 Quality conformance inspection shall consist of the groups A and B examination and tests specified in 4.3.3 and 4.3.4. The samples shall be subjected to tests in the order shown.

4.3.3 Group A examination and tests. - Group A examination and tests shall consist of those specified in table III. Sampling shall be in accordance with MIL-STD-105, inspection level II. The Acceptable Quality Level (A.Q.L.) shall be 1.5 percent.

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Table III - Group A examination and tests.

Tests	Requirement paragraph	Test paragraph
Visual, dimensional, and mechanical examination	3.4.1 thru 3.4.4, 3.7, 3.8 and 3.9	4.4.3
Acoustical quality	3.5.10	4.4.7
Dielectric strength	3.6.3	4.4.5

4.3.4 Group B tests. - The group B tests shall consist of the following:

Tests	Requirement paragraph	Test paragraph
Frequency response	3.5.6	4.4.4
Power handling	3.5.5	4.4.4

4.3.4.1 Sampling for group B tests. - Sampling for group B tests shall be as specified in table IV:

Table IV - Sampling for Group B tests.

Lot size number of units	Sample size Units to be tested	Acceptance number	Rejection number
Up to 17	2	0	1
18 to 35	3	0	1
36 to 70	4	0	1
71 to 140	5	0	1
141 to 280	7	0	1
281 to 560	10	1	2
561 to 1120	14	1	2
1121 to 2240	20	2	3
2241 to 4480	28	3	4
4481 to 8960	40	4	5

4.3.5 Rejected lots. - If an inspection lot is rejected, the supplier may replace it with a new lot, rework it or screen out defectives, and again inspect it, using the next larger sample size. Reinspected lots shall be kept separate from new lots, and shall be clearly identified as reinspected lots.

4.4 Test procedures. -

4.4.1 Test conditions. - Unless otherwise specified herein, all measurements and tests shall be made in accordance with the following:

(a) A. F. input - Across the 600-ohm tap of the matching transformer.
(b) Loudspeaker and microphone mounted under free field conditions.
(c) Test microphone - 5 feet on axis of speaker.
(d) Audio level - Maximum efficiency.
(e) Ambient temperature - Between 20° and 30° centigrade (C.).
(f) Relative humidity - Below 50 percent.
(g) Reference sound pressure level - The reference sound pressure level shall be 0.002 dynes per centimeter.

4.4.2 Test equipment and inspection facilities. - Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required quality conformance inspection. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

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4.4.3 Visual, dimensional, and mechanical examination. - Loudspeakers shall be examined visually, dimensionally, and mechanically to determine conformance with this specification.

4.4.4 Electrical tests. - Tests shall be conducted to determine conformance with the requirements of 3.5.5, 3.5.6 and 3.5.9.

4.4.5 Dielectric test. - The dielectric withstanding voltage test shall be performed in accordance with method 301 (200 ± 5 volts at 60 c.p.s.) of MIL-STD-202. Dielectric withstanding voltage shall be measured between all circuit conductors and between all circuit conductors and the metal parts of the loudspeaker. Care should be taken not to apply the test voltage across the loudspeaker voice coil or across the primary or secondary windings of the audio transformer.

4.4.6 Salt spray. - The loudspeaker assembly shall be subjected to a salt spray test as specified in method 101, condition B, of Standard MIL-STD-202. Following the salt spray test the requirements of 3.6.6 shall apply.

4.4.7 Acoustic quality. - A constant voltage having an r.m.s. value to provide rated power input (see 3.5.5) shall be applied to the voice coil terminals and the frequency varied continuously from 200 to 5,000 c.p.s. The output of the loudspeaker shall meet the requirements of 3.5.10.

4.4.8 Harmonic distortion. - A harmonic distortion test shall be performed on the loudspeaker assembly. A constant voltage having an r.m.s. value to provide rated power input (see 3.5.5) at 400 c.p.s. shall be applied to the primary of the a.f. transformer at frequencies of 400, 2,000 and 5,000 c.p.s. ± 10 percent. The harmonic distortion in the acoustic output of the loudspeaker at each frequency shall be determined by using a distortion analyzer or equivalent. The total harmonic distortion at each frequency shall be determined by the following formula:

$$\text{Percent r.m.s. distortion} = \frac{P_2^2 + P_3^2 + \ldots . P_n^2 \times 100}{P_1^2 + P_2^2 + P_3^2 + \ldots . P_n^2}$$

Where P_1 is the pressure amplitude of the fundamental and P_2 P_3, and higher are the pressure amplitudes of the harmonic components in the output. The harmonic distortion in the acoustic output of the loudspeaker at each frequency shall not exceed the limit specified in 3.5.8.

4.4.9 Shock and vibration tests. - The loudspeaker assembly shall be tested in accordance with grade A, class I of MIL-S-901, while mounted as shown on figure 5 with the bulkhead mounting flanges attached to the loudspeaker assembly (see Drawing RE49F767). The loudspeaker assembly shall also be tested for vibration in accordance with Standard MIL-STD-167 (see 3.6.2).

4.4.10 Temperature cycling. - The loudspeaker shall be subjected to the temperature cycling test, method 102, of Standard MIL-STD-202.

4.4.11 Moisture resistance. - The loudspeaker shall be subjected to the moisture resistance test in accordance with method 106, condition B, of Standard MIL-STD-202, with the exception of step 7.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Loudspeaker. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or

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method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1) in accordance with MIL-E-17555.

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation, packaging, packing, and marking. - The equipment, accessories, and technical publications shall be preserved and packaged level A or C, packed level A, B, or C and marked as specified (see 6.1) in accordance with MIL-E-17555. There shall be one loudspeaker per package.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation, packaging, packing, or marking requirements if other than those required by 5.1 (see 5.2).

6.2 Preproduction. -

6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. - THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5965-N016SH)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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